Year 1 AS Pure Maths

Chapter 8

The binomial expansion

Practice Booklet

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8.1 Pascal's triangle

Practice Book, printed page 62.

1. Use Pascal's triangle to write down the expansion of:

a. $(3-x)^5$

b. $(2x+y)^4$

c. $(3x-2y)^5$

Hint

The coefficients in the expansion of $(a+b)^n$ are in the $(n+1)$th row of Pascal's triangle.

2. Find the coefficient of x^3 in the expansion of:

a. $(5+2x)^4$

b. $(2-3x)^5$

c. $(3-\frac{1}{2}x)^4$

Hint

The term in x^n is the $(n+1)$th term in the expansion in ascending powers of x. When substituting a term with two or more characters, such as $2x$, into the expansion of $(a+b)^n$ use brackets to make sure you don't make a mistake.

3. Fully expand the expression $(1-2x)(1+3x)^3$.

Hint

Expand $(1+3x)^3$, then multiply each term by 1 and by $-2x$.

4. E

a. Find all the terms of the binomial expansion of $(2+3x)^4$. Give each term in its simplest form. (2 marks)

b. Write down the expansion of $(2-3x)^4$ in ascending powers of x, giving each term in its simplest form. (1 mark)

5. E/P The coefficient of x^2 in the expansion of $(3+kx)^4$ is 1350. Find two possible values of the constant, k. (3 marks)

6. E/P

a. Expand $(4+k)^3$. (2 marks)

b. Hence, or otherwise, write down the expansion of $(4+x^2-x)^3$ in ascending powers of x. (3 marks)

7. E/P The coefficient of x^2 in the expansion of $(5+x)(2-kx)^3$ is 18. Find two possible values of the constant, k. (3 marks)

8.2 Factorial notation

Practice Book, printed page 62–63.

1. Without using a calculator, work out:

a. $0!5!$

b. $\frac{12!}{9!}$

c. $\frac{20!}{18!}$

Hint

By definition, $0! = 1$. For parts **b** and **c**, you can simplify by cancelling. For example, $12!$ can be written as $12 \times 11 \times 10 \times 9!$

2. Use a calculator to work out:

a. $\binom{16}{9}$

b. ${}^{12}C_5$

c. $\binom{22}{10}$

d. ${}^{18}C_{11}$

Hint

${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$, $n, r \in \mathbb{N}$.
You can use the ${}^nC_r$ and ! functions on your calculator.

3. Write each value, p, q, r and s, from Pascal's triangle using ${}^nC_r$ notation.

$$\begin{array}{ccccccccccc} & & & & & 1 & & & & & \\ & & & & 1 & & 1 & & & & \\ & & & 1 & & 2 & & 1 & & & \\ & & 1 & & 3 & & 3 & & 1 & & \\ & 1 & & 4 & & p & & 4 & & 1 & \\ 1 & & 5 & & 10 & & q & & 5 & & 1 \\ 1 & & 6 & & 15 & & r & & s & & 6 & & 1 \end{array}$$

4. Write the 8th number in the 15th row of Pascal's triangle in ${}^nC_r$ form.

Hint

The rth entry in the nth row of Pascal's triangle is given by ${}^{n-1}C_{r-1}$ or $\binom{n-1}{r-1}$.

5. The probability of rolling n threes when a fair dice is rolled 10 times is given by

$$\binom{10}{n}\left(\frac{1}{6}\right)^n\left(\frac{5}{6}\right)^{10-n}, \quad 0 \leqslant n \leqslant 10$$

Find the probability of rolling:

a. no threes

b. 5 threes

6. E Given that $\binom{25}{11} = \frac{25!}{11!p!}$, write down the value of p. (1 mark)

7. E Given that $\binom{30}{q} = \frac{30!}{q!8!}$, write down the value of q. (1 mark)

8. E/P

a. Show that ${}^{k+2}C_k = \frac{(k+2)(k+1)}{2}$ (3 marks)

b. Given that ${}^{k+2}C_k = 36$, find the value of k. (3 marks)

c. Explain why there is only one solution for k in part **b**. (1 mark)

8.3 The binomial expansion

Practice Book, printed page 63–64.

1. Use the binomial theorem to find the expansion of:

a. $(1+x)^5$

b. $(2+x)^6$

c. $(10+x)^4$

2. Use the binomial theorem to find the expansion of:

a. $(3-2x)^4$

b. $(4+\frac{1}{2}x)^5$

c. $(x-2)^6$

Hint

The binomial expansion is:

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n \quad (n \in \mathbb{N})$$

where ${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$ $(n, r \in \mathbb{N})$

3. Use the binomial theorem to find the first four terms, in ascending powers of x, in the expansion of:

a. $(1+x)^{10}$

b. $(1+x)^{15}$

c. $(2+x)^9$

Hint

You need to expand each expression up to the term in x^3.

4. Use the binomial theorem to find the first four terms, in ascending powers of x, in the expansion of:

a. $(1-\frac{x}{2})^{10}$

b. $(2-\frac{1}{3}x)^9$

c. $(3+2x)^8$

Hint

Use brackets when you substitute terms into the binomial expansion formula, and include any negative signs.
For example, the 4th term of $(2y-\frac{x}{2})^5$ is

$$\binom{5}{3}(2y)^2\left(-\frac{x}{2}\right)^3 = 10(4y^2)\left(-\frac{x^3}{8}\right) = -5y^2x^3$$

5. E Find the first four terms, in ascending powers of x, of the binomial expansion of $(2-3x)^5$, giving each term in its simplest form. (4 marks)

6. E Given that $(3-x)^{12} \equiv A + Bx + Cx^2 + \ldots$, find the values of the integers A, B and C. (4 marks)

7. E Find the first four terms, in ascending powers of x, of the binomial expansion of $(1-\frac{2}{3}x)^9$. (4 marks)

8. E/P Find the binomial expansion of $(x-\frac{2}{x})^5$, giving each term in its simplest form. (4 marks)

8.4 Solving binomial problems

Practice Book, printed page 64–65.

1. Find the coefficient of x^3 in the expansion of:

a. $(1+x)^{14}$

b. $(2+x)^{10}$

c. $(5-x)^7$

2. Find the coefficient of x^4 in the expansion of:

a. $(3-2x)^9$

b. $(5+3x)^7$

c. $(\frac{1}{2}+8x)^{12}$

Hint

The general term in the expansion of $(a+b)^n$ is $\binom{n}{r}a^{n-r}b^r$.

3. Find the coefficient of x^3 in the expansion of $(3-x)(2+3x)^6$.

Hint

First expand $(2+3x)^6$ up to the term in x^3, then multiply by $(3-x)$. There are two ways of making a term in x^3: (constant term $\times$ x^3 term) and (x term $\times$ x^2 term).

4. E Find the coefficient of x^3 in the expansion of $(1+\frac{2}{3}x)^6$. (3 marks)

5. E/P The coefficient of x^2 in the expansion of $(1+2x)^n$ is 60. Given that $n>0$, find the value of n. (3 marks)

6. P The coefficient of x^2 in the expansion of $(3+ax)^5$ is 4320. Find two possible values of the constant, a.

7. P The coefficient of x^3 in the expansion of $(2-bx)^6$ is -20. Find the value of the constant, b.

8. E/P

a. Find the first four terms, in ascending powers of x, of the binomial expansion of $(2-3x)^8$, giving each term in its simplest form. (4 marks)

b. Hence, or otherwise, find the coefficient of x^3 in the expansion of $(1+\frac{x}{2})(2-3x)^8$. (3 marks)

9. E/P

a. Find the first three terms, in ascending powers of x, of the binomial expansion of $(3+px)^6$, where p is a non-zero constant. Give each term in its simplest form. (4 marks)

b. Given that, in this expansion, the coefficient of x^2 is 5 times the coefficient of x, find the value of p. (2 marks)

8.5 Binomial estimation

Practice Book, printed page 65–66.

1.

a. Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 - \frac{x}{5})^9$. Give each term in its simplest form.

b. Use your expansion to estimate the value of 0.985^9, giving your answer to 4 decimal places.

Hint

For part **b**, work out the value of x to substitute into the expansion by letting $1 - \frac{x}{5} = 0.985$.

2.

a. Find the first three terms, in ascending powers of x, of the binomial expansion of $(1 - 2x)^6$. Give each term in its simplest form.

b. If x is so small that x^3 and higher powers can be ignored, show that $(3 + x)(1 - 2x)^6 \approx 3 - 35x + 168x^2$.

Hint

Multiply your answer to part **a** by $(3 + x)$ and collect like terms.

3.

a. Write down the first four terms, in ascending powers of x, of the binomial expansion of $(3 + \frac{x}{4})^8$.

b. By substituting an appropriate value for x, find an approximate value for 3.05^8.

Hint

For part **b**, work out the value of x to substitute into the expansion by letting $\frac{x}{4} = 0.05$.

4. E

a. Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + \frac{x}{4})^9$. Give each term in its simplest form. (4 marks)

b. Use your expansion to estimate the value of 1.015^9, giving your answer to 3 decimal places. (3 marks)

5. E/P

a. Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 - 3x)^6$. Give each term in its simplest form. (4 marks)

b. If x is small, so that x^3 and higher powers can be ignored, show that $(1 + \frac{x}{2})(1 - 3x)^6 \approx 1 - \frac{35x}{2} + 126x^2$. (2 marks)

6. E/P

a. Find the first three terms, in ascending powers of x, of the binomial expansion of $(2 - \frac{x}{2})^7$, giving each term in its simplest form. (4 marks)

b. Explain how you would use your expansion to give an estimate for the value of 1.985^7. (1 mark)

7. E/P

a. Find the first four terms, in ascending powers of x, of the binomial expansion of $(3 - \frac{x}{5})^{10}$. Give each term in its simplest form. (4 marks)

b. Use your answer to part **a** to estimate the value of 2.995^{10}, giving your answer to 6 decimal places. (3 marks)

c. Use your calculator to evaluate 2.995^{10} and calculate the percentage error in your answer to part **b**. (2 marks)

Problem set A

Bronze

In the binomial expansion of $(3-kx)^6$, where k is a constant, the coefficient of x^3 is -4320. Calculate:

a. the value of k (4 marks)

b. the value of the coefficient of x^4 in the expansion. (2 marks)

Silver

a. Write down the first three terms, in ascending powers of x, of the binomial expansion of $(2-px)^{11}$, where p is a non-zero constant, giving each term in its simplest form. (4 marks)

b. Given that, in the expansion of $(2-px)^{11}$, the coefficient of x is q and the coefficient of x^2 is $10q$, find the value of p and the value of q. (3 marks)

Gold

In the binomial expansion of $(3+\frac{x}{k})^8$, the coefficient of x^2 is 3 times the coefficient of x^3. Find the value of the constant, k. (7 marks)

Problem set B

Bronze

$f(x)=(2+kx)(1-5x)^4$, where k is a constant. The expansion, in ascending powers of x, of $f(x)$ up to and including the term in x^2 is $A-37x+Bx^2$, where A and B are constants.

a. Find the first three terms, in ascending powers of x, of the binomial expansion of $(1-5x)^4$, giving each term in its simplest form. (4 marks)

b. Write down the value of A. (1 mark)

c. Find the value of k. (2 marks)

d. Hence find the value of B. (2 marks)

Silver

$f(x)=(1+kx)(2-5x)^4$, where k is a constant. The expansion, in ascending powers of x, of $f(x)$ up to and including the term in x^2 is $A-64x+Bx^2$, where A and B are constants. Find the values of A, B and k. (7 marks)

Gold

$f(x)=(3+px)(1+qx)^5$, where p and q are positive constants. The expansion, in ascending powers of x, of $f(x)$ up to and including the term in x^3 is $3+17x+\frac{70}{3}x^2+kx^3$, where k is a constant.

a. Find the value of p and the value of q. (7 marks)

b. Hence find the value of k. (1 mark)

Selected exam question bank questions

Question 9

In the binomial expansion of $(1 + 3x)^n$, where $n > 0$, the coefficient of x^2 is 6 times the coefficient of x. Work out the value of n. (4 marks)

Question 20

a. Expand $(1 + \frac{2}{x})^2$, simplifying each term. (2 marks)

b. Use the binomial expansion to find, in ascending powers of x, the first four terms in the expansion of $(1 + \frac{2}{5}x)^6$, simplifying each term. (4 marks)

c. Hence find the coefficient of x in the expansion of $(1 + \frac{2}{x})^2(1 + \frac{2}{5}x)^6$. (2 marks)

Question 29

a. Find the first 4 terms, in ascending powers of p, of the binomial expansion of $(q + p)^{100}$. (3 marks)

Let p represent the probability that any one lightbulb in a batch of 100 is faulty and let q represent the probability it is not faulty. Given that $p = 0.02$,

b. use your answer from part **a** to work out the probability that no more than 3 lightbulbs in a batch of 100 are faulty. Give your answer correct to 5 decimal places. (5 marks)

Question 40

a. Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2 - \frac{x}{15})^{10}$. Give each term in its simplest form. (4 marks)

$f(x) = (p + qx)(2 - \frac{x}{15})^{10}$, where p and q are constants. Given that the first two terms, in ascending powers of x, in the series expansion of $f(x)$ are 512 and $-\frac{256x}{3}$,

b. find the value of p (2 marks)

c. find the value of q. (2 marks)

Question 51

a. Fully expand $(1 + x)^5$ (3 marks)

b. Use your answer to part **a** to show that:

i. $(1 + \sqrt{5})^5 = 176 + 80\sqrt{5}$

ii. $\log_2(1 + \sqrt{5})^5 = p + \log_2(11 + 5\sqrt{5})$ where p is an integer.

(6 marks)

Question 61

a. Find the binomial expansion of $(1 + 3x)^4$. (4 marks)

b. Hence show that $(1 + 3x)^4 + (1 - 3x)^4 = 2 + 108x^2 + 162x^4$ (3 marks)

c. Use your answer to part **b** to show that the curve with equation $f(x) = (1 + 3x)^4 + (1 - 3x)^4$ has only one stationary point and find the coordinates. (4 marks)

Answers

8.1 Pascal's triangle

1.
- **a.** $243 - 405x + 270x^2 - 90x^3 + 15x^4 - x^5$
- **b.** $y^4 + 8xy^3 + 24x^2y^2 + 32x^3y + 16x^4$
- **c.** $243x^5 - 810x^4y + 1080x^3y^2 - 720x^2y^3 + 240xy^4 - 32y^5$

2.
- **a.** 160
- **b.** -1080
- **c.** $-\frac{3}{2}$

3. $1 + 7x + 9x^2 - 27x^3 - 54x^4$

4.
- **a.** $16 + 96x + 216x^2 + 216x^3 + 81x^4$
- **b.** $16 - 96x + 216x^2 - 216x^3 + 81x^4$

5. $k = \pm 5$

6.
- **a.** $64 + 48k + 12k^2 + k^3$
- **b.** $64 - 48x + 60x^2 - 25x^3 + 15x^4 - 3x^5 + x^6$

7. $k = 1$ or $k = -\frac{3}{5}$

8.2 Factorial notation

1.
- **a.** 120
- **b.** 1320
- **c.** 380

2.
- **a.** 11440
- **b.** 792
- **c.** 646646
- **d.** 31824

3. $p = {}^4C_2$, $q = {}^5C_3$, $r = {}^6C_3$, $s = {}^6C_4$

4. ${}^{15}C_7$

5.
- **a.** 0.162 (3 s.f.)
- **b.** 0.0130 (3 s.f.)

6. $p = 14$

7. $q = 22$

8.
- **a.** ${}^{k+2}C_k = \dfrac{(k+2)!}{k!(k+2-k)!} = \dfrac{(k+1)(k+2)}{2}$
- **b.** 7
- **c.** k is a positive integer, $k > 0$.

8.3 The binomial expansion

1.
- **a.** $1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$
- **b.** $64 + 192x + 240x^2 + 160x^3 + 60x^4 + 12x^5 + x^6$
- **c.** $10000 + 4000x + 600x^2 + 40x^3 + x^4$

2.
- **a.** $81 - 216x + 216x^2 - 96x^3 + 16x^4$
- **b.** $1024 + 640x + 160x^2 + 20x^3 + \frac{5}{4}x^4 + \frac{1}{32}x^5$
- **c.** $64 - 192x + 240x^2 - 160x^3 + 60x^4 - 12x^5 + x^6$

3.
- **a.** $1 + 10x + 45x^2 + 120x^3$
- **b.** $1 + 15x + 105x^2 + 455x^3$
- **c.** $512 + 2304x + 4608x^2 + 5376x^3$

4.
- **a.** $1 - 5x + \frac{45}{4}x^2 - 15x^3$
- **b.** $512 - 768x + 512x^2 - \frac{1792}{9}x^3$
- **c.** $6561 + 34992x + 81648x^2 + 108864x^3$

5. $32 - 240x + 720x^2 - 1080x^3$

6. $A = 531441$, $B = -2125764$, $C = 3897234$

7. $1 - 6x + 16x^2 - \frac{224}{9}x^3$

8. $x^5 - 10x^3 + 40x - \frac{80}{x} + \frac{80}{x^3} - \frac{32}{x^5}$

8.4 Solving binomial problems

1.
- **a.** 364
- **b.** 15360
- **c.** -21875

2.
- **a.** 489888
- **b.** 354375
- **c.** 7920

3. 10800

4. $\frac{160}{27}$

5. $n = 6$

6. $a = \pm 4$

7. 0.5

8.
- **a.** $256 - 3072x + 16128x^2 - 48384x^3$
- **b.** -40320

9.
- **a.** $729 + 1458px + 1215p^2x^2$
- **b.** $p = 6$

8.5 Binomial estimation

1.
- **a.** $1 - \frac{9}{5}x + \frac{36}{25}x^2 - \frac{84}{125}x^3$
- **b.** 0.8728

2.
- **a.** $1 - 12x + 60x^2$
- **b.** Expand $(3 + x)(1 - 2x)^6$ up to terms in x^2.

3.
- **a.** $6561 + 4374x + \frac{5103}{4}x^2 + \frac{1701}{8}x^3$
- **b.** Using $x = 0.2$, 7488.531

4.
- **a.** $1 + \frac{9}{4}x + \frac{9}{4}x^2 + \frac{21}{16}x^3$
- **b.** Using $x = 0.06$, $(1.015)^9 = 1.143$ (3 d.p.)

5.
- **a.** $1 - 18x + 135x^2 - 540x^3$
- **b.** Expand $(1 + \frac{x}{2})(1 - 3x)^6$ up to terms in x^2.

6.
- **a.** $128 - 224x + 168x^2$
- **b.** Substitute $x = 0.03$ in the expansion.

7.
- **a.** $59049 - 39366x + \frac{59049}{5}x^2 - \frac{52488}{25}x^3$
- **b.** 58072.198320
- **c.** $1.64 \times 10^{-7}\%$ (3 s.f.)

Problem set A

Bronze

- **a.** $k = 2$
- **b.** 2160

Silver

- **a.** $2048 - 11264px + 28160p^2x^2$
- **b.** $p = -4$, $q = 45056$

Gold $k = 2$

Problem set B

Bronze

- **a.** $1 - 20x + 150x^2$
- **b.** $A = 2$
- **c.** $k = 3$
- **d.** $B = 240$

Silver $A = 16$, $k = 6$, $B = -360$

Gold

- **a.** $p = 12$, $q = \frac{1}{3}$
- **b.** $k = \frac{130}{9}$

Selected exam question bank answers

Question 9 $n = 5$

Question 20

a. $1 + \frac{4}{x} + \frac{4}{x^2}$

b. $1 + \frac{12x}{5} + \frac{12x^2}{5} + \frac{32x^3}{25}$

c. $\frac{428}{25}$

Question 29

a. $q^{100} + 100q^{99}p + 4950q^{98}p^2 + 161700q^{97}p^3$

b. 0.85896

Question 40

a. $1024 - \frac{1024}{3}x + \frac{256}{5}x^2$

b. $p = \frac{1}{2}$

c. $q = \frac{1}{12}$

Question 51

a. $1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$

b.

i. $(1 + \sqrt{5})^5 = 1 + 5\sqrt{5} + 10(5) + 10(5\sqrt{5}) + 5(25) + (25\sqrt{5})$
$= 176 + 80\sqrt{5}$

ii. $\log_2(1 + \sqrt{5})^5 = \log_2(176 + 80\sqrt{5}) = \log_2 16(11 + 5\sqrt{5})$
$= \log_2 16 + \log_2(11 + 5\sqrt{5}) = 4 + \log_2(11 + 5\sqrt{5})$

Question 61

a. $1 + 12x + 54x^2 + 108x^3 + 81x^4$

b. Expand $(1 - 3x)^4$ in a similar way to $(1 + 3x)^4$ and add the two expansions. The x and x^3 terms cancel out.

c. $(0, 2)$